

Evidence Bundle 2 — Linux Log Analysis

1. Evidence of recent system-log review using journalctl.

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(moysoreobasa@kali)~$ sudo journalctl --since "today"
[sudo] password for moysoreobasa:
Oct 01 18:52:35 kali kernel: Linux version 6.12.25-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-9) 14.2.0, GNU ld (GNU Binutils for Debian) 2.44)
Oct 01 18:52:35 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.25-amd64 root=UUID=c344711b-0575-4503-9228-9ebd452e9c92 ro quiet splash
Oct 01 18:52:35 kali kernel: BIOS-provided physical RAM map:
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000009fbff] usable
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000000009fc00-0x0000000000000fffff] reserved
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x000000000000f0000-0x0000000000000fffff] reserved
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000000100000-0x000000000000dfffff] usable
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000000dffff000-0x00000000000dffffffffff] ACPI data
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00ffff] reserved
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00ffff] reserved
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000ffff00000-0x00000000ffffffffff] reserved
Oct 01 18:52:35 kali kernel: BIOS-e820: [mem 0x00000000100000000-0x000000001acccfffff] usable
Oct 01 18:52:35 kali kernel: NX (Execute Disable) protection: active
Oct 01 18:52:35 kali kernel: APIC: Static calls initialized
Oct 01 18:52:35 kali kernel: SMBIOS 2.5 present.
Oct 01 18:52:35 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Oct 01 18:52:35 kali kernel: DMI: Memory slots populated: 0/0
Oct 01 18:52:35 kali kernel: Hypervisor detected: KVM
Oct 01 18:52:35 kali kernel: kvm-clock: Using mrs 4b564d01 and 4b564d00
Oct 01 18:52:35 kali kernel: kvm-clock: using sched offset of 16401663592 cycles
Oct 01 18:52:35 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
Oct 01 18:52:35 kali kernel: tsc: Detected 2693.764 MHz processor
Oct 01 18:52:35 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable => reserved
Oct 01 18:52:35 kali kernel: e820: remove [mem 0x00000000-0x00000fff] usable
Oct 01 18:52:35 kali kernel: last_pfn = 0x1accd00 max_arch_pfn = 0x400000000
Oct 01 18:52:35 kali kernel: MTRR map: 6 entries (3 fixed + 3 variable; max 35), built from 16 variable MTRRs
Oct 01 18:52:35 kali kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 01 18:52:35 kali kernel: e820: update [mem 0x00000000-0xffffffff] usable => reserved
Oct 01 18:52:35 kali kernel: last_pfn = 0xe0000 max_arch_pfn = 0x400000000
Oct 01 18:52:35 kali kernel: Found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
```

2. Authentication or privilege-use evidence showing relevant event or confirming that no significant event was observed.

```
File Actions Edit View Help
(moysoreobasa@kali)~$ sudo journalctl --since "today" | grep -i "sudo"
[sudo] password for moysoreobasa:
Oct 01 18:56:25 kali sudo[9047]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/systemctl status auditd
Oct 01 18:56:25 kali sudo[9047]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 18:56:25 kali sudo[9047]: pam_unix(sudo:session): session closed for user root
Oct 01 18:56:43 kali sudo[9871]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/apt update
Oct 01 18:56:43 kali sudo[9871]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 18:56:38 kali sudo[9871]: pam_unix(sudo:session): session closed for user root
Oct 01 18:58:38 kali sudo[13383]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/apt install auditd audispd-plugins
Oct 01 18:58:38 kali sudo[13383]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 18:58:38 kali sudo[13383]: pam_unix(sudo:session): session closed for user root
Oct 01 18:59:56 kali sudo[15961]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/systemctl start auditd
Oct 01 18:59:56 kali sudo[15961]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 18:59:56 kali sudo[15961]: pam_unix(sudo:session): session closed for user root
Oct 01 18:59:56 kali sudo[15986]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/systemctl enable auditd
Oct 01 18:59:56 kali sudo[15986]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:00:19 kali sudo[17261]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/systemctl status auditd
Oct 01 19:00:19 kali sudo[17261]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:00:19 kali sudo[17261]: pam_unix(sudo:session): session closed for user root
Oct 01 19:02:55 kali sudo[22078]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/nano /etc/audit/rules.d/custom.rules
Oct 01 19:02:55 kali sudo[22078]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:07:38 kali sudo[22078]: pam_unix(sudo:session): session closed for user root
Oct 01 19:07:40 kali sudo[34455]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/journalctl _COMM=sshd
Oct 01 19:07:40 kali sudo[34455]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:10:33 kali sudo[34455]: pam_unix(sudo:session): session closed for user root
Oct 01 19:10:51 kali sudo[36538]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/nano /etc/audit/rules.d/custom.rules
Oct 01 19:10:51 kali sudo[36538]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:13:34 kali sudo[36538]: pam_unix(sudo:session): session closed for user root
Oct 01 19:14:57 kali sudo[44607]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/systemctl restart auditd
Oct 01 19:14:57 kali sudo[44607]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:14:57 kali sudo[44607]: pam_unix(sudo:session): session closed for user root
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File Actions Edit View Help
Oct 01 19:25:15 kali sudo[63488]: pam_unix(sudo:session): session closed for user root
Oct 01 19:25:39 kali sudo[64192]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/ausearch -k program_execution
Oct 01 19:25:39 kali sudo[64192]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:26:16 kali sudo[64192]: pam_unix(sudo:session): session closed for user root
Oct 01 19:26:16 kali sudo[65259]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/ausearch -k auth_failures
Oct 01 19:26:16 kali sudo[65259]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:26:17 kali sudo[65259]: pam_unix(sudo:session): session closed for user root
Oct 01 19:26:27 kali sudo[65598]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/aureport
Oct 01 19:26:27 kali sudo[65598]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:26:28 kali sudo[65598]: pam_unix(sudo:session): session closed for user root
Oct 01 19:26:28 kali sudo[65623]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/aureport --failed
Oct 01 19:26:28 kali sudo[65623]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:26:28 kali sudo[65623]: pam_unix(sudo:session): session closed for user root
Oct 01 19:26:28 kali sudo[65663]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/aureport --login
Oct 01 19:26:28 kali sudo[65663]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:26:29 kali sudo[65663]: pam_unix(sudo:session): session closed for user root
Oct 01 19:40:07 kali sudo[91447]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/sbin/ausearch -k passwd_changes
Oct 01 19:40:07 kali sudo[91447]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 19:40:08 kali sudo[91447]: pam_unix(sudo:session): session closed for user root
Oct 01 19:59:15 kali sudo[118979]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 01 19:59:15 kali sudo[118979]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 20:11:03 kali sudo[118979]: pam_unix(sudo:session): session closed for user root
Oct 01 20:11:25 kali sudo[137184]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 01 20:11:25 kali sudo[137184]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 20:11:27 kali sudo[137184]: pam_unix(sudo:session): session closed for user root
Oct 01 20:13:44 kali sudo[140479]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 01 20:13:44 kali sudo[140479]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
Oct 01 20:13:46 kali sudo[140479]: pam_unix(sudo:session): session closed for user root
Oct 01 21:05:29 kali sudo[13318]: moysoreobasa : TTY=pts/0 ; PWD=/home/moysoreobasa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 01 21:05:29 kali sudo[13318]: pam_unix(sudo:session): session opened for user root(uid=0) by moysoreobasa(uid=1001)
```

3. Error/warning evidence from syslog or an equivalent source.

```
(moyosoreobasa@kali)-[~]
$ sudo journalctl -p err --since "today"
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
Oct 01 18:52:35 kali (sd-exec[578]): /usr/lib/systemd/system-generators/systemd-sshd-generator terminated by signal ABRT.
Oct 01 18:53:54 kali lightdm[3872]: gkr-pam: unable to locate daemon control file
Oct 01 18:53:55 kali lightdm[1112]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Oct 01 18:54:20 kali obexd[4835]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 18:54:20 kali obexd[4835]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 19:00:03 kali (sd-exec[16440]): /usr/lib/systemd/system-generators/systemd-sshd-generator terminated by signal SEGV.
Oct 01 19:38:44 kali xfce4-screensaver-dialog[88379]: pam_winbind(xfce4-screensaver:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM
Oct 01 19:38:44 kali xfce4-screensaver-dialog[88379]: pam_winbind(xfce4-screensaver:auth): internal module error (retval = PAM_AUTHINFO_UNAVAILABLE(9), user = 'moyosore
-- Boot a203e13038f8468d98123665bc55343b --
Oct 01 20:59:08 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
Oct 01 20:59:08 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 01 20:59:08 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
Oct 01 21:03:17 kali lightdm[8497]: gkr-pam: unable to locate daemon control file
Oct 01 21:03:18 kali lightdm[1027]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Oct 01 21:03:40 kali obexd[9866]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 21:03:40 kali obexd[9866]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 21:03:40 kali obexd[9866]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
lines 1-19/19 (END)

File Actions Edit View Help
(moyosoreobasa@kali)-[~]
$ sudo journalctl -p warning --since "today"
Oct 01 18:52:35 kali kernel: acpi PNP0A03:00: fail to add MMCONFIG information, can't access extended configuration space under this bridge
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 01 18:52:35 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
Oct 01 18:52:35 kali kernel: synth probe
Oct 01 18:52:35 kali kernel: clocksource: Long readout interval, skipping watchdog check: cs_nsec: 1056702312 wd_nsec: 1056701974
Oct 01 18:52:35 kali systemd-sshd-generator: Configuration directory '/etc/ssh/' does not exist! No units generated.
Oct 01 18:52:35 kali systemd-sysv-generator[593]: SysV service '/etc/init.d/stunnel4' lacks a native systemd unit file, automatically generating a unit file for comp
Oct 01 18:52:35 kali systemd-sysv-generator[593]: Please update package to include a native systemd unit file.
Oct 01 18:52:35 kali systemd-sysv-generator[593]: ! This compatibility logic is deprecated, expect removal soon. !
Oct 01 18:52:35 kali systemd-sysv-generator[593]: SysV service '/etc/init.d/dnsm2tcp' lacks a native systemd unit file, automatically generating a unit file for compa
Oct 01 18:52:35 kali systemd-sysv-generator[593]: Please update package to include a native systemd unit file.
Oct 01 18:52:35 kali systemd-sysv-generator[593]: ! This compatibility logic is deprecated, expect removal soon. !
Oct 01 18:52:35 kali systemd-sysv-generator[593]: SysV service '/etc/init.d/inetssm' lacks a native systemd unit file, automatically generating a unit file for compa
Oct 01 18:52:35 kali systemd-sysv-generator[593]: Please update package to include a native systemd unit file.
Oct 01 18:52:35 kali systemd-sysv-generator[593]: ! This compatibility logic is deprecated, expect removal soon. !
Oct 01 18:52:35 kali (sd-exec[578]): /usr/lib/systemd/system-generators/systemd-sshd-generator terminated by signal ABRT.
Oct 01 18:52:35 kali systemd-journal[616]: File /var/log/journal/9e7e68b7dd044db0898e2067ebd41cab/system.journal corrupted or uncleanly shut down, renaming and repl
Oct 01 18:52:38 kali (haveged)[775]: haveged.service: Referenced but unset environment variable evaluates to an empty string: DAEMON_ARGS
Oct 01 18:52:39 kali accounts-daemon[781]: failed to check if user 'aceofdiamond' in cache dir is present on system: No such file or directory
Oct 01 18:52:46 kali wireplumber[1172]: spa.bluez5: BlueZ system service is not available
Oct 01 18:52:46 kali wireplumber[1172]: spa.bluez5: BlueZ system service is not available
Oct 01 18:52:56 kali accounts-daemon[781]: failed to check if user 'aceofdiamond' in cache dir is present on system: No such file or directory
Oct 01 18:53:54 kali lightdm[3872]: gkr-pam: unable to locate daemon control file
Oct 01 18:53:55 kali lightdm[1112]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Oct 01 18:53:58 kali wireplumber[4135]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Oct 01 18:53:58 kali wireplumber[4135]: spa.bluez5: BlueZ system service is not available
Oct 01 18:54:08 kali accounts-daemon[781]: failed to check if user 'aceofdiamond' in cache dir is present on system: No such file or directory
Oct 01 18:54:20 kali obexd[4835]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 18:54:20 kali obexd[4835]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-reg
Oct 01 18:54:24 kali colord[4725]: failed to get edid data: EDID length is too small
```

4. A table containing Event/Condition, Evidence Source, Time, User/Service, Security Significance and Recommended Action.

Event	Evidence source	Time	User	Security significance	Recommended action
Privileged sudo activity was recorded, including commands used to manage auditd and edit audit rules	journalctl --since today grep -i sudo	18:56–19:14, Oct. 1	User moyosoreobasa executed as root	Provides an audit trail of privileged administrative activity and supports accountability for security-control changes	Review privileged activity periodically and retain sudo logs for audit and investigation
Failed system/service operations were recorded, including accounts-daemon, Docker and UPower-related failures	journalctl --since today grep -i failed	18:52–18:58, Oct. 1	accounts-daemon, dockerd, UPower	Repeated failures may indicate service/configuration issues that could affect system reliability or security monitoring	Investigate recurring service failures and determine whether they are expected in the lab environment
Warning events included deprecated SysV service	journalctl -p warning --since today	18:52 onward, Oct. 1	systemd-sysv-generator, wireplumber,	Shows conditions requiring monitoring, although the output	Determine which warnings are expected for the Kali VM and document or remediate

compatibility and several desktop/service warnings			lightdm	alone does not establish a security incident	persistent issues
SSH generator and journal-related errors were recorded	journalctl -p err --since "today"	18:52–18:53, Oct. 1	systemd-ssh-generator, systemd-journal	Errors involving logging or SSH-related services can affect availability or the completeness of security monitoring evidence	Verify service configuration and confirm that required security logging remains operational

The journal review identified both privileged administrative activity and several system/service errors and warnings. The sudo records provide traceability for security related administrative commands, including changes to the auditd configuration. Error and warning records included vmwgfx, systemd, accounts-daemon, Docker, UPower and desktop-service issues. These events demonstrate the value of centralized log monitoring because they provide evidence of administrative activity and system conditions that may require investigation. However, the presence of an error or warning does not by itself establish a security incident; events should be correlated with the relevant service, user activity and system context.